

Board Investment Report

Automated Micro-Fulfilment Centre — Dubai, United Arab Emirates

Topic 9 — AI Capital-Budgeting Dashboard | Prepared for the Capital Expenditure Committee,
NovaRetail GCC | 03 August 2026

Entity: NovaRetail GCC

Horizon: 6-Year DCF

WACC: 11.50%

Tax: UAE 9%

Recommendation: APPROVE

TOTAL CAPITAL OUTLAY

AED 24.00M

CapEx 22.0M + working capital 2.0M

NET PRESENT VALUE

AED 12.08M

AED 12,083,628 at 11.50%

INTERNAL RATE OF RETURN

26.30%

14.80 pts above hurdle

PAYBACK PERIOD

3.10 Yrs

Discounted: 3.98 Yrs

Academic notice. NovaRetail GCC is a **hypothetical entity constructed for academic evaluation**. It is not a real company, and no figure in this report constitutes investment advice or a vendor quotation. The corporate-finance methodology, the UAE tax treatment (9% federal corporate tax) and the DEWA tariff references are real and current; the capital-cost estimates and the operating benefit case are illustrative. Section 12 sets out the full provenance and the model's limitations without embellishment.

Investment proposal	Automated Micro-Fulfilment Centre (goods-to-person robotics, ASRS, warehouse control system), urban Dubai
Sponsoring entity	NovaRetail GCC — omnichannel retail group, United Arab Emirates (hypothetical)
Decision requested	Approval of the full AED 24,000,000 commitment (Option 1 of four)
Evaluation horizon	6 years, plus terminal salvage and working-capital recovery in Year 6
Discount rate	11.50% weighted average cost of capital (derivation in Section 4)
Analytical engine	CapExIQ deterministic TypeScript capital-budgeting engine; AI advisory layer is non-computational
Document status	Board memorandum — for decision

1. EXECUTIVE SUMMARY AND RECOMMENDATION

Management asks the Committee to approve a capital commitment of **AED 24,000,000** to build an Automated Micro-Fulfilment Centre in Dubai. The commitment comprises **AED 22,000,000** of capital expenditure and **AED 2,000,000** of incremental working capital, the latter recovered in full at the end of Year 6. Evaluated over a six-year life at a weighted average cost of capital of **11.50%**, the proposal generates a net present value of **AED 12,083,628** on a present value of inflows of **AED 36,083,628**.

Recommendation: APPROVE the full AED 24,000,000 investment (Option 1), subject to two conditions.

1. **Release capital against measured Year-1 savings, not against the calendar.** Benefit realisation dominates every other variable in the model: a $\pm 20\%$ movement in operating benefits swings NPV by **AED 16.67 million**, roughly twice the effect of the next-ranked driver. Capital should be drawn in tranches unlocked by verified savings run-rates rather than by elapsed time.
2. **Require a vendor performance guarantee and a secondary-market equipment buyback before commitment.** The pessimistic scenario is genuinely value-destroying — NPV (**AED 4,940,625**) with an IRR of 8.23%, below the cost of capital. A contractual floor under both throughput performance and residual value converts that downside from an unhedged loss into a bounded one.

1.1 Why the Committee should approve

- **The return is well clear of the hurdle.** IRR of 26.30% exceeds the 11.50% WACC by 14.80 percentage points. Modified IRR of 19.34%, which reinvests interim cash at the cost of capital rather than at the project's own return, still clears the hurdle by 7.84 points.
- **Capital efficiency is strong.** A profitability index of 1.5035 means each AED 1.00 committed returns AED 1.5035 of present value — AED 0.5035 of value created per dirham deployed. Simple return on investment over the life is 123.3%.
- **Capital is recovered inside the asset's life with margin.** Simple payback is 3.10 years and discounted payback 3.98 years, against a six-year evaluation horizon.
- **The downside is survivable and quantified.** Operating benefits can fall **29.0%** below plan before NPV reaches zero, and total outlay can overrun by **50.4%** (to AED 36.08 million) before value is destroyed. A 5,000-run Monte Carlo simulation puts the probability of a negative NPV at approximately **0.3%**.
- **The probability-weighted outcome remains positive.** Weighting the optimistic, base and pessimistic scenarios 50/25/25 gives an expected NPV of **AED 9,560,152**.

1.2 What the Committee should watch

One automated risk alert fires on the base case: **downside exposure in the pessimistic scenario**. Under a combined stress of +15% capital cost, -25% benefits, +15% operating cost and a 14.5% discount rate, the project destroys AED 4.94 million of value. The engine's benefit-shortfall rule (which fires when the tolerable shortfall falls below 15%) and its salvage-dependence rule (which fires when salvage exceeds 15% of NPV) do *not* fire: the tolerance is 29.0% and the present value of salvage is 8.61% of NPV. Those two rules are reported here as satisfied, not as active alerts.

2. THE INVESTMENT DECISION AND THE STRATEGIC OPTIONS

NovaRetail GCC's manual fulfilment operation has become the binding constraint on its online growth in Dubai and Abu Dhabi. The question before the Committee is not whether automation is directionally correct but *how much capital to commit, and when*. Four options were framed. Only Option 1 is modelled end-to-end by the deterministic engine; the remaining three are assessed qualitatively against it, and no net present value is quoted for them, because none has been run through the same schedule.

Option	Year-0 commitment	Case for	Case against
1. Approve in full RECOMMENDED	AED 24,000,000	Captures the full benefit curve from Year 1; the only configuration valued end-to-end, at NPV AED 12,083,628, IRR 26.30%, PI 1.5035.	Commits the entire sum before any savings are observed — which is precisely what condition (1) of the recommendation is designed to mitigate.
2. Phased build	AED 14,000,000 then AED 10,000,000	Caps initial exposure at AED 14.0 million and buys a genuine abandonment option after the first phase.	Defers a material share of the benefit stream and adds re-mobilisation cost. Not modelled on the same basis, so its NPV is not comparable and is deliberately not quoted.
3. Delay 12 months	Nil in Year 0	Buys another year of demand evidence and possibly cheaper robotics.	Forfeits roughly a full year of the benefit stream while the fulfilment bottleneck persists; the value of the delay option was not quantified by the engine.
4. Reject	Nil	Preserves capital for alternative uses.	Forgoes AED 12.08 million of modelled value creation and leaves the operational constraint unaddressed.

A note on comparability. It would be straightforward, and misleading, to present indicative NPVs for Options 2 to 4. This report does not, because those figures would not come from the same verified schedule as Option 1. Where a number is not produced by the engine, it is not stated.

3. CORPORATE FINANCE METHODOLOGY

The appraisal follows standard discounted cash-flow practice. Cash flows are incremental, after tax, and stated in nominal AED; financing cash flows are excluded from the flows themselves and captured instead through the discount rate. Each measure below is computed by the deterministic engine from the same Section 6 schedule.

3.1 Free cash flow

$$FCF = (EBITDA - \text{Depreciation}) \times (1 - t) + \text{Depreciation} - \Delta \text{Working Capital} - \text{CapEx} + \text{Salvage}$$

The cash the asset actually releases each year. Depreciation is subtracted to obtain taxable profit, then added back because it is not a cash outflow — its only cash effect is the tax it shelters.

3.2 Net present value

$$NPV = \sum [FCF_t / (1 + r)^n] - \text{Initial Outlay}, \quad r = 11.50\%$$

The value added to the firm today, in dirhams. A positive NPV means the project earns more than the capital it consumes costs. This is the primary decision rule.

3.3 Internal rate of return

$$IRR = r \text{ such that } NPV(r) = 0$$

The annualised return implied by the cash-flow profile. It is compared with the 11.50% hurdle; it is a ranking check, not a substitute for NPV.

3.4 Modified internal rate of return

$$MIRR = (\text{FV of inflows reinvested at } r / \text{PV of outflows at } r)^{(1/n)} - 1$$

Corrects the IRR's implicit assumption that interim cash is reinvested at the project's own return; MIRR reinvests it at the cost of capital instead, which is the honest assumption.

3.5 Profitability index

$$PI = \text{PV of future inflows} / \text{Initial Outlay} = 36,083,628 / 24,000,000 = 1.5035$$

Value created per dirham committed. It is the right ranking metric when capital is rationed, which it is.

3.6 Payback and discounted payback

$$\text{Payback} = \text{years until cumulative FCF} \geq 0 \quad | \quad \text{Discounted payback} = \text{years until cumulative PV} \geq 0$$

Liquidity and exposure measures, read directly off the cumulative rows of the Section 6 schedule by linear interpolation within the crossing year. Neither measures profitability; both measure how long capital is at risk.

Convention. Year 0 is the commitment date. All operating flows are treated as occurring at each year end. Depreciation is straight-line *to salvage value*, not to zero: $(\text{AED } 22,000,000 - \text{AED } 2,000,000) \div 6 \text{ years} = \text{AED } 3,333,333 \text{ per year}$, generating an annual tax shield of **AED 300,000** at the 9% rate.

4. COST OF CAPITAL — WACC DERIVATION

The 11.50% discount rate is built up from observable market inputs and two explicit judgemental premia, both of which are disclosed rather than buried in a rounded rate.

4.1 Cost of equity

Component	Rate	Basis
Risk-free rate	4.20%	UAE sovereign benchmark yield
Equity risk premium × beta (1.15 × 6.00%)	6.90%	Market ERP geared to retail-logistics beta
UAE country risk premium	0.75%	Sovereign spread adjustment
Project execution premium	3.50%	First-of-kind automation delivery risk
Cost of equity (K_e)	15.35%	4.20% + (1.15 × 6.00%) + 0.75% + 3.50%

4.2 Cost of debt

Component	Rate	Basis
3-month EIBOR (live)	3.79%	Prevailing UAE interbank benchmark
Credit spread	2.50%	Corporate borrowing margin
Pre-tax cost of debt	6.29%	3.79% + 2.50%
After-tax cost of debt (K_d)	5.72%	6.29% × (1 – 0.09)

4.3 Weighted average

$$\begin{aligned}
 \text{WACC} &= (E/V \times K_e) + (D/V \times K_d) \\
 &= (0.60 \times 15.35\%) + (0.40 \times 5.72\%) \\
 &= 9.21\% + 2.29\% \\
 &= 11.50\%
 \end{aligned}$$

The target structure is 60% equity / 40% debt. The interest tax shield is captured in the after-tax cost of debt at the UAE statutory 9% rate, so it is not double-counted in the cash flows. The resulting **11.50%** is the hurdle applied throughout this report and the discount rate underlying every factor in Section 6.

5. CAPITAL OUTLAY BREAKDOWN

Component	Amount (AED)	Share	Treatment
Automation equipment	18,000,000	75.0%	Capitalised; depreciated straight-line to salvage
Installation and systems integration	2,500,000	10.4%	Capitalised into the depreciable base
Software and licensing	1,200,000	5.0%	Capitalised into the depreciable base
Training and launch support	300,000	1.3%	Capitalised into the depreciable base
Total capital expenditure	22,000,000	91.7%	Depreciable base for Section 6
Incremental working capital	2,000,000	8.3%	Not depreciated; recovered in full in Year 6
Total Year-0 outlay	24,000,000	100.0%	Denominator for PI, ROI and payback

Shares are computed on the AED 24,000,000 total outlay and rounded to one decimal place; they sum to 100.0%.

5.1 Depreciation and the terminal year

- **Depreciable base:** AED 22,000,000 (capital expenditure only; working capital is never depreciated).
- **Method:** straight-line to salvage — $(22,000,000 - 2,000,000) \div 6 = \text{AED } 3,333,333 \text{ per year}$.
- **Annual tax shield:** $3,333,333 \times 9\% = \text{AED } 300,000 \text{ per year}$.
- **Year 6 terminal inflows:** salvage AED 2,000,000 plus working-capital recovery AED 2,000,000 = AED 4,000,000, on top of that year's operating cash flow of AED 9,186,330.

Because the asset is written down to its salvage value rather than to zero, there is no book gain or loss on disposal in Year 6 and therefore no balancing tax charge. The salvage inflow is recognised gross.

6. SIX-YEAR FREE CASH FLOW SCHEDULE

All figures in AED. Negative amounts are shown in parentheses. Year-1 operating savings of AED 7,500,000 grow at 4% per annum; Year-1 contribution margin of AED 2,500,000 grows at 5%; Year-1 operating expenses of AED 2,200,000 grow at 3%. Tax is charged at the UAE statutory 9% on EBIT. Depreciation is straight-line to salvage at AED 3,333,333 per year for six years. Cash flows are discounted at the 11.50% WACC derived in Section 4.

Line item	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Operating savings	—	7,500,000	7,800,000	8,112,000	8,436,480	8,773,939	9,124,897
Contribution margin	—	2,500,000	2,625,000	2,756,250	2,894,063	3,038,766	3,190,704
Total benefits	—	10,000,000	10,425,000	10,868,250	11,330,543	11,812,705	12,315,601
Less: operating expenses	—	(2,200,000)	(2,266,000)	(2,333,980)	(2,403,999)	(2,476,119)	(2,550,403)
EBITDA	—	7,800,000	8,159,000	8,534,270	8,926,543	9,336,585	9,765,198
Less: depreciation	—	(3,333,333)	(3,333,333)	(3,333,333)	(3,333,333)	(3,333,333)	(3,333,333)
EBIT	—	4,466,667	4,825,667	5,200,937	5,593,210	6,003,252	6,431,864
Less: tax @ 9%	—	(402,000)	(434,310)	(468,084)	(503,389)	(540,293)	(578,868)
NOPAT	—	4,064,667	4,391,357	4,732,852	5,089,821	5,462,959	5,852,997
Add back: depreciation	—	3,333,333	3,333,333	3,333,333	3,333,333	3,333,333	3,333,333
Operating cash flow	—	7,398,000	7,724,690	8,066,186	8,423,154	8,796,293	9,186,330
Salvage value	—	—	—	—	—	—	2,000,000
Working capital recovery	—	—	—	—	—	—	2,000,000
Initial outlay	(24,000,000)	—	—	—	—	—	—
Free cash flow	(24,000,000)	7,398,000	7,724,690	8,066,186	8,423,154	8,796,293	13,186,330
Discount factor @ 11.50%	1.0000	0.8969	0.8044	0.7214	0.6470	0.5803	0.5204
Present value of FCF	(24,000,000)	6,634,978	6,213,429	5,818,936	5,449,734	5,104,172	6,862,380
Cumulative FCF	(24,000,000)	(16,602,000)	(8,877,310)	(811,124)	7,612,030	16,408,323	29,594,653
Cumulative discounted FCF	(24,000,000)	(17,365,022)	(11,151,594)	(5,332,657)	117,077	5,221,249	12,083,628

Rows are rendered directly from the engine's schedule data; they are not transcribed by hand. Present values are computed at full precision ($1 \div 1.115^n$) and then rounded to the nearest dirham; the discount factors are displayed to four decimal places, so re-computing a present value from the displayed factor will differ by a few hundred dirhams. Summing the rounded present-value row gives AED 12,083,629 against the stated NPV of AED 12,083,628 — a one-dirham rounding difference, not a discrepancy.

6.1 Reading the schedule

- **The NPV ties out.** The cumulative discounted row closes at **AED 12,083,628** in Year 6 — the same figure headlined on the cover and in Section 7. The sum of the present values of Years 1 to 6 is AED 36,083,628, from which the AED 24,000,000 outlay is deducted.
- **Payback falls in Year 4.** Cumulative undiscounted FCF turns positive between Year 3 (AED (811,124)) and Year 4 (AED 7,612,030), giving 3.10 years by interpolation. Cumulative discounted FCF turns positive in the same year (AED (5,332,657) to AED 117,077), giving 3.98 years — the tighter, and the one the Committee should use.
- **Year 6 is inflated by the terminal items.** Free cash flow of AED 13,186,330 in Year 6 includes AED 2,000,000 of salvage and AED 2,000,000 of working-capital recovery. Stripping both leaves AED

9,186,330 of genuine operating cash flow, so the year is not carried by its terminal value.

- **The tax shield is small but real.** Depreciation of AED 3,333,333 shelters AED 300,000 of tax each year; over six years that is AED 1,800,000 of undiscounted cash retained.

7. INVESTMENT METRICS AGAINST BENCHMARKS

Metric	Base case	Benchmark	Verdict	Reading
Net present value	AED 12,083,628	> AED 0	PASS	Value created after paying for all capital consumed
Internal rate of return	26.30%	> 11.50% WACC	PASS	14.80 points of spread over the hurdle
Modified IRR	19.34%	> 11.50% WACC	PASS	Holds up when interim cash is reinvested at the cost of capital
Profitability index	1.5035	> 1.00	PASS	AED 1.5035 of present value per AED 1.00 committed
Simple payback	3.10 years	< 4.0 years	PASS	Capital recovered during Year 4 of a 6-year life
Discounted payback	3.98 years	< 4.5 years	PASS	Recovered in Year 4 even after the cost of capital
Return on investment	123.3%	> 0%	PASS	Undiscounted; see derivation note below
Present value of inflows	AED 36,083,628	> AED 24,000,000	PASS	Covers the outlay 1.50 times over

Derivation note: return on investment is the undiscounted net cash gain over the life divided by the total outlay — AED 29,594,653 (the Year-6 cumulative free cash flow in Section 6) ÷ AED 24,000,000 = 123.3%. It is reported for completeness only; it ignores the time value of money and must not be read alongside the IRR as if the two were comparable rates. Every benchmark in the third column is NovaRetail's standing capital-approval threshold, not a figure produced by the model.

All eight tests pass. No metric in the base case falls short of its threshold. That is a statement about the base case only; Sections 8 to 11 test what happens when the base case is wrong.

8. SCENARIO ANALYSIS

Three internally consistent states of the world were run end-to-end through the engine. Each flexes capital cost, benefits, operating cost and the discount rate together, because in practice these move together rather than independently.

Scenario	Assumption set	NPV (AED)	IRR	PI	Payback	Decision
Optimistic	CapEx ×0.95, benefits ×1.10, OpEx ×0.95, WACC 10.5%	19,013,977	33.59%	1.830	2.63 yrs	APPROVE
Base	As modelled in Section 6, WACC 11.50%	12,083,628	26.30%	1.504	3.10 yrs	APPROVE
Pessimistic	CapEx ×1.15, benefits ×0.75, OpEx ×1.15, WACC 14.5%	(4,940,625)	8.23%	0.819	5.06 yrs	REJECT

Expected NPV, weighted 50% optimistic / 25% base / 25% pessimistic: AED 9,560,152. The probability-weighted outcome is comfortably positive, but the pessimistic branch is not a rounding error — it destroys AED 4.94 million and returns 8.23%, below the 11.50% cost of capital, with a payback that runs past the asset's economic life. This single result is the one automated risk alert firing on the base case, and it is the reason condition (2) of the recommendation asks for a vendor performance guarantee and an equipment buyback.

9. SENSITIVITY ANALYSIS

Normalisation. Every driver below is flexed by an identical $\pm 20\%$ about its base value, one at a time, with all other drivers held at base. This matters: a tornado chart built from driver-specific ranges (say $\pm 10\%$ on capital cost but $\pm 30\%$ on demand) ranks the analyst's assumptions rather than the project's economics. Applying one common percentage makes the swing column a like-for-like measure of how much NPV each driver actually controls.

Rank	Driver	NPV at -20% (AED M)	NPV at +20% (AED M)	Swing (AED M)	Direction
1	Operating benefits	3.75	20.42	16.67	Positive
2	Project life	7.76	16.15	8.39	Positive
3	Capital expenditure	16.21	7.96	8.25	Inverse
4	Discount rate (WACC)	14.81	9.64	5.17	Inverse
5	Operating expenses	13.87	10.30	3.57	Inverse
6	Savings growth rate	—	—	1.10	Positive
7	Salvage value	—	—	0.37	Positive

Base-case NPV is AED 12.08 million. For ranks 6 and 7 the engine recorded the swing magnitude only; the individual endpoints are therefore shown as unavailable rather than reconstructed, since a symmetric back-calculation would be an assumption, not a result.

9.1 What the ranking means for the decision

Operating benefits swing NPV by AED 16.67 million — almost exactly twice the second-ranked driver and larger than the base-case NPV itself. Benefit realisation is not one risk among several; it is the risk. Capital cost discipline (rank 3) and financing cost (rank 4) matter, but a project team that delivers the savings can absorb a 20% capital overrun and still return AED 7.96 million, whereas a team that delivers the asset on budget but misses the savings by 20% returns AED 3.75 million. Salvage value is close to irrelevant at AED 0.37 million of swing, which is a healthy sign: the case does not depend on the resale market.

9.2 Break-even frontiers

Variable	Break-even point	Interpretation
Operating benefits	-29.0%	Benefits may fall 29.0% below plan before NPV reaches zero
Total capital outlay	+50.4%	Outlay may overrun to AED 36.08 million before value is destroyed
Discount rate	26.30%	NPV is zero at a 26.30% discount rate — by definition the IRR
Operating expenses	+136%	Running costs could more than double before the case fails
Salvage dependence	8.61%	Present value of salvage is 8.61% of NPV — well inside the 15% alert threshold

The 29.0% benefit tolerance sits comfortably above the engine's 15% alert threshold, so the benefit-shortfall rule does not fire. It is nonetheless the tightest of the four operating frontiers, which is consistent with the tornado ranking and with condition (1) of the recommendation.

10. MONTE CARLO RISK SIMULATION

Scenario analysis tests three futures; the simulation tests five thousand. Four inputs were allowed to vary simultaneously, each with a distribution chosen to match how that input actually behaves, and the full six-year schedule was re-solved on every draw.

Parameter	Distribution	Rationale
Capital expenditure	Triangular	Bounded above and below by vendor quotation ranges; overruns are asymmetric
Operating savings	Normal	Aggregate of many independent process improvements
Operating expenses	Triangular	Bounded by contracted maintenance and energy tariffs
Discount rate (WACC)	Normal	Driven by EIBOR movement around a central expectation

ITERATIONS

5,000

Random seed 12345

MEAN NPV

~AED 10.5M

Below the AED 12.08M base case

P(NPV < 0)

~0.3%

Roughly 1 run in 333

P(NPV > 0)

~99.7%

Complement of the above

Two readings matter. First, the mean simulated NPV of approximately **AED 10.5 million** sits *below* the AED 12,083,628 base case. That gap is informative: the distribution is left-skewed, because capital overruns and benefit shortfalls hurt more than the symmetric upside helps. The base case is therefore mildly optimistic relative to the probability-weighted centre of the distribution, and the Committee should anchor on the AED 10.5 million figure rather than on the headline.

Second, a **0.3%** probability of a negative NPV is a statement about the assumed distributions, not about the world. It says the project is robust to the ordinary co-movement of costs, savings and rates that the model contemplates. It says nothing about a vendor insolvency, a technology failure or a demand collapse — none of which is in the distribution. The pessimistic scenario in Section 8, which does destroy value, is the appropriate reference point for those risks, and the run seed (12345) is fixed so that this figure is exactly reproducible.

11. RISK REGISTER AND MITIGATIONS

Automated alerts firing on the base case: one. The engine's rule set evaluates three conditions. Only *downside exposure in the pessimistic scenario* is triggered. The benefit-shortfall rule (fires below a 15% tolerance) and the salvage-dependence rule (fires above a 15% share of NPV) are satisfied at 29.0% and 8.61% respectively and are **not** active alerts. They are listed in the register below as monitored risks because they are real exposures, not because the engine flagged them.

Ref	Risk	Status	Exposure	Mitigation
R1	Downside exposure — pessimistic scenario	ALERT ACTIVE	NPV (4,940,625)	Vendor performance guarantee with liquidated damages tied to throughput; secondary-market equipment buyback agreed before commitment; stage-gate abandonment right after Phase 1 acceptance.
R2	Benefit realisation shortfall	Monitored — not firing	29.0% tolerance	Release capital against measured Year-1 savings run-rate rather than the calendar; monthly cost-per-order reporting to the Committee against the AED 7,500,000 Year-1 target.
R3	Capital cost overrun	Monitored	50.4% tolerance	Fixed-price integration contract; contingency drawn only on Committee approval; overrun ceiling reported against the AED 36.08 million break-even.
R4	Financing cost movement (EIBOR)	Monitored	5.17M NPV swing	Fix or cap the debt tranche at drawdown; re-run the WACC derivation in Section 4 at each stage gate using the then-current 3-month EIBOR.
R5	Asset life shorter than six years	Monitored	8.39M NPV swing	Preventive maintenance contract for the full term; obsolescence review at Year 3; modular refresh path specified in the procurement scope.
R6	Salvage / residual value dependence	Satisfied — not firing	8.61% of NPV	No action required. The case does not rest on resale; the buyback in R1 provides a floor in any event.
R7	Operating cost inflation	Monitored	+136% tolerance	Energy consumption metered against the DEWA tariff assumption; maintenance costs contracted at fixed escalation.
R8	Model risk — unmodelled tax and ramp-up effects	Disclosed	Not quantified	Section 12 lists each simplification. Re-run the schedule with a benefit ramp-up curve and the AED 375,000 zero-rate band before financial close.

12. DATA PROVENANCE AND MODEL LIMITATIONS

The Committee is entitled to know exactly which numbers are observed and which are assumed. The following is a complete statement, including the parts that weaken the case.

12.1 What is genuine external data

Source	What it is	How it is used
DataCo Smart Supply Chain Mendeley Data 8gx2fvg2k6, CC BY 4.0	Genuine external operational benchmark dataset covering US supply-chain transactions	Operational benchmarking and dashboard illustration only. It does NOT feed the financial model. No figure in Section 6 is derived from it. It is US data applied to a UAE context, and it is presented as context, not as evidence for the benefit case.
DEWA commercial tariffs	Real and current published Dubai electricity tariffs	Energy cost component of the operating expense line
UAE Federal Corporate Tax	Real and current — 9% statutory rate, effective June 2023	Tax line in the Section 6 schedule
3-month EIBOR	Live UAE interbank benchmark, 3.79%	Base rate in the cost-of-debt build-up in Section 4

12.2 What is estimated

- **Capital cost figures are illustrative estimates, not vendor quotations.** The AED 18,000,000 equipment line, the AED 2,500,000 installation line and the software and training lines have not been tendered. A real tender is the single most valuable piece of evidence the Committee could obtain before financial close.
- **The benefit case is assumed, not observed.** Year-1 savings of AED 7,500,000 and contribution margin of AED 2,500,000, and their 4% and 5% growth rates, are management estimates. Section 9 shows this is the dominant driver, so the weakest-evidenced input is also the most important one.
- **Salvage value of AED 2,000,000** is an estimate of six-year-old automation residual value in a thin secondary market.

12.3 Known simplifications in the model

- **Flat 9% tax.** The model applies 9% to the whole of EBIT and ignores the UAE **AED 375,000 zero-rate band**. This overstates the tax charge by up to AED 33,750 per year and so understates NPV very slightly — a conservative error, but an error.
- **No loss carry-forward.** Every year in the base case is profitable, so the omission is immaterial here; in the pessimistic scenario it is not, and the AED (4,940,625) result should be read as somewhat worse than a full tax model would produce.
- **No benefit ramp-up curve.** Year-1 savings are assumed to be realised in full from the first year. Real automation programmes ramp over two to four quarters. This is the most optimistic single assumption in the model, and it biases NPV upward.

- **Year-end cash flow convention.** All operating flows are treated as arriving at year end, which understates present value slightly relative to mid-year timing.
- **No inflation differential.** Savings, margin and cost lines each grow at a single fixed rate; no separate general-inflation adjustment is applied.
- **No terminal or continuing value.** The asset is assumed to be worth only its salvage value after Year 6, with no continuing operation. This is conservative.

Net direction of the biases. The absent ramp-up curve pushes NPV up; the ignored zero-rate band, the year-end convention and the absence of any continuing value push it down. They have not been netted, and the Committee should not assume they cancel.

13. ARTIFICIAL INTELLIGENCE GOVERNANCE STATEMENT

CapExIQ is an AI-assisted capital-budgeting dashboard. The Committee should be precise about what the AI does and does not do in this report.

13.1 The mathematics is deterministic

Every figure in Sections 6 to 11 — the free cash flow schedule, NPV, IRR, MIRR, profitability index, payback, the scenario set, the sensitivity swings, the break-even frontiers and the Monte Carlo distribution — is produced by **deterministic TypeScript** running fixed formulae over fixed inputs. No language model computes, adjusts, rounds or selects any number in this document. Given the same inputs the engine returns the same outputs, every time; the Monte Carlo section is seeded (12345) precisely so that even the stochastic result is reproducible.

13.2 The AI layer is advisory and non-computational

The AI assistant drafts narrative, explains methodology in plain language, and answers questions about results it is given. It has no write access to the financial engine, cannot alter an assumption, cannot approve or reject a project, and cannot originate a figure. Where this report contains judgement — the ranking of options, the two conditions attached to the recommendation, the reading of the risk register — that judgement is management's and is open to challenge on its merits.

13.3 Accountability

Final responsibility for this investment decision rests with the Chief Financial Officer and the Capital Expenditure Committee — not with the AI system. CapExIQ is a decision-support tool. It structures evidence, enforces arithmetic consistency and makes assumptions explicit. It does not carry fiduciary responsibility, and no part of this report should be read as delegating that responsibility to software.

13.4 Auditability

- All inputs, formulae and outputs are held in version-controlled source; this report is regenerated from that source by `scripts/generate-board-report.js` rather than assembled by hand.
- The Section 6 schedule is rendered from structured data, so a figure cannot drift between the schedule and the headline metrics without the source changing.
- Assumptions, data sources, methodology and known limitations are documented separately in the repository (`ASSUMPTIONS.md`, `DATA_SOURCES.md`, `FINANCIAL_METHODODOLOGY.md`, `MODEL_LIMITATIONS.md`, `AI_GOVERNANCE.md`).

14. APPROVAL

The Capital Expenditure Committee is asked to approve **Option 1 — the full AED 24,000,000 commitment** to the Automated Micro-Fulfilment Centre, subject to the two conditions set out in Section 1: tranche release against measured Year-1 savings, and a vendor performance guarantee with a secondary-market equipment buyback agreed before commitment.

CHIEF FINANCIAL OFFICER

Signature

Name / Date

CHIEF EXECUTIVE OFFICER

Signature

Name / Date

CAPITAL EXPENDITURE
COMMITTEE CHAIR

Signature

Name / Date

Document control. CapExIQ Board Investment Report — Topic 9, AI Capital-Budgeting Dashboard | Entity: NovaRetail GCC (hypothetical, for academic evaluation) | Base case at WACC 11.50%, six-year horizon | Generated 03 August 2026 by `scripts/generate-board-report.js` from the deterministic CapExIQ engine | Supersedes all prior ad-hoc versions of this document, including any five-year Section 6 schedule.